

HW #4 - Equations of Lines

Date _____ Period _____

Find the slope of the line through each pair of points.

1) $(-16, 2), (17, -2)$

2) $(-15, -2), (-15, -12)$

3) $(3, 8), (-13, 4)$

4) $(-9, 13), (-13, 13)$

Write the slope-intercept form of the equation of the line through the given point with the given slope.

5) through: $(-4, 0)$, slope = 2

6) through: $(-5, 5)$, slope = $-\frac{6}{5}$

Write the slope-intercept form of the equation of the line through the given points.

7) through: $(-1, -4)$ and $(2, 2)$

8) through: $(-5, -1)$ and $(-5, 1)$

Write the slope-intercept form of the equation of each line given the slope and y-intercept.

9) Slope = $-\frac{9}{2}$, y-intercept = 5

10) Slope = $\frac{2}{3}$, y-intercept = 3

Write the slope-intercept form of the equation of each line.

11) $x = 1$

12) $7x - 6y = -30$

Write the standard form of the equation of each line given the slope and y-intercept.

13) Slope = -2, y-intercept = 3

14) Slope = $-\frac{1}{2}$, y-intercept = -1

Write the slope-intercept form of the equation of the line described.

15) through: $(5, -5)$, parallel to $x = 0$

16) through: $(-4, -4)$, parallel to $y = \frac{1}{4}x - 2$

17) through: $(1, 4)$, parallel to $y = 7x$

18) through: $(2, 2)$, perp. to $y = 4$

19) through: $(-4, -4)$, perp. to $y = -x - 1$

20) through: $(-2, 5)$, perp. to $y = \frac{2}{9}x - 3$

Write the standard form of the equation of the line described.

21) through: $(-4, 3)$, parallel to $y = 4x - 5$

22) through: $(4, 0)$, parallel to $y = -\frac{1}{4}x + 2$

23) through: $(-4, -2)$, perp. to $y = -\frac{7}{6}x + 2$

24) through: $(1, 4)$, perp. to $y = 0$

Answers to HW #4 - Equations of Lines

1) $-\frac{4}{33}$

5) $y = 2x + 8$

9) $y = -\frac{9}{2}x + 5$

13) $2x + y = 3$

17) $y = 7x - 3$

21) $4x - y = -19$

2) Undefined

6) $y = -\frac{6}{5}x - 1$

10) $y = \frac{2}{3}x + 3$

14) $x + 2y = -2$

18) $x = 2$

22) $x + 4y = 4$

3) $\frac{1}{4}$

7) $y = 2x - 2$

11) $x = 1$

15) $x = 5$

19) $y = x$

23) $6x - 7y = -10$

4) 0

8) $x = -5$

12) $y = \frac{7}{6}x + 5$

16) $y = \frac{1}{4}x - 3$

20) $y = -\frac{9}{2}x - 4$

24) $x = 1$